Heart Disease Modeling: Logistic Regression and Random Forests

Project Overview

This project uses R to examine relationships between patient health indicators and the presence of heart disease. Logistic regression and classification random forest models were developed to predict heart-disease status, while a regression random forest model was used to predict maximum heart rate achieved. The models were evaluated and compared to identify important predictors and assess their predictive performance.

Variables used in the analysis:

Variable	**What does it represent?**
age	The person's age in years
sex	The person's sex (1 = male, 0 = female)
cp	The type of chest pain experienced (0=no pain, 1=typical angina, 2=atypical angina, 3=non-anginal pain)
trestbps	The person's resting blood pressure
chol	The person's cholesterol measurement in mg/dl
fbs	The person's fasting blood sugar is greater than 120 mg/dl (1 = true, 0 = false)
restecg	Resting electrocardiographic measurement (0=normal, 1=having ST-T wave abnormality, 2=showing probable or definite left ventricular hypertrophy by Estes' criteria)
thalach	The person's maximum heart rate achieved
exang	Exercise-induced angina (1=yes, 0=no)
oldpeak	ST depression induced by exercise relative to rest ('ST' relates to positions on the ECG plot)
slope	The slope of the peak exercise ST segment (1=upsloping, 2=flat, 3=downsloping)
ca	The number of major vessels (0-3)
target	Heart disease (0=no, 1=yes)

Install Libraries

In [22]:
```
print("This step will first install three R packages. Please wait until the pa
ckages are fully installed.")
print("Once the installation is complete, this step will print 'Installation c
omplete!'")

install.packages("ResourceSelection")
install.packages("pROC")
install.packages("rpart.plot")

print("Installation complete!")
```

```
[1] "This step will first install three R packages. Please wait until the pac
kages are fully installed."
[1] "Once the installation is complete, this step will print 'Installation co
mplete!'"
```

```
[1] "Installation complete!"
```

Prepare The Dataset

In [23]:
```
heart_data <- read.csv(file="heart_disease.csv", header=TRUE, sep=",")

# Converting appropriate variables to factors
heart_data <- within(heart_data, {
   target <- factor(target)
   sex <- factor(sex)
   cp <- factor(cp)
   fbs <- factor(fbs)
   restecg <- factor(restecg)
   exang <- factor(exang)
   slope <- factor(slope)
   ca <- factor(ca)
   thal <- factor(thal)
})

head(heart_data, 10)

print("Number of variables")
ncol(heart_data)

print("Number of rows")
nrow(heart_data)
```

A data.frame: 10 × 14

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	th
<int>	<fct>	<fct>	<int>	<int>	<fct>	<fct>	<int>	<fct>	<dbl>	<fct>	<fct>	<
62	1	2	130	231	0	1	146	0	1.8	1	3	3
58	0	0	130	197	0	1	131	0	0.6	1	0	2
60	0	3	150	240	0	1	171	0	0.9	2	0	2
63	1	0	140	187	0	0	144	1	4.0	2	2	3
62	1	0	120	267	0	1	99	1	1.8	1	2	3
63	0	2	135	252	0	0	172	0	0.0	2	0	2
43	1	0	150	247	0	1	171	0	1.5	2	0	2
42	1	2	120	240	1	1	194	0	0.8	0	0	3
59	1	2	126	218	1	1	134	0	2.2	1	1	1
48	1	0	124	274	0	0	166	0	0.5	1	0	3

```
[1] "Number of variables"
```

14

```
[1] "Number of rows"
```

303

Model #1 - First Logistic Regression Model

Logistic regression model for heart disease *(target)* using the variables age *(age)*, resting blood pressure *(trestbps)*, exercised induced angina (exang) and maximum heart rate achieved *(thalach)*.

In [3]:
```
# Create the complete model
logit1 <- glm(target ~ age + trestbps + exang + thalach , data = heart_data, f
amily = "binomial")

summary(logit1)
```

```
Call:
glm(formula = target ~ age + trestbps + exang + thalach, family = "binomial",
    data = heart_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0935  -0.7944   0.4954   0.8133   2.2343

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.021121   1.784194  -0.572   0.5671
age         -0.017549   0.017144  -1.024   0.3060
trestbps    -0.014888   0.008337  -1.786   0.0741 .
exang1      -1.624981   0.305774  -5.314 1.07e-07 ***
thalach      0.031095   0.007275   4.274 1.92e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 417.64  on 302  degrees of freedom
Residual deviance: 323.14  on 298  degrees of freedom
AIC: 333.14

Number of Fisher Scoring iterations: 4
```

In [5]:
```
library(ResourceSelection)
print("Hosmer-Lemeshow Goodness of Fit Test")
hl = hoslem.test(logit1$y, fitted(logit1), g=50)
hl
```

```
ResourceSelection 0.3-6          2023-06-27

[1] "Hosmer-Lemeshow Goodness of Fit Test"

        Hosmer and Lemeshow goodness of fit (GOF) test

data:  logit1$y, fitted(logit1)
X-squared = 44.622, df = 48, p-value = 0.612
```

Wald's test - z-values with their corresponding p-values

In [11]:
```
default_model_data <- heart_data[c('age', 'trestbps', 'exang', 'thalach')]
pred <- predict(logit1, newdata=default_model_data, type='response')

depvar_pred = as.factor(ifelse(pred >= 0.5, '1', '0'))

# Create the confusion matrix
conf.matrix <- table(heart_data$target, depvar_pred)[c('0','1'),c('0','1')]
rownames(conf.matrix) <- paste("Actual", rownames(conf.matrix), sep = ": targe
t=")
colnames(conf.matrix) <- paste("Prediction", colnames(conf.matrix), sep = ": t
arget=")

# Print nicely formatted confusion matrix
print("Confusion Matrix")
format(conf.matrix,justify="centre",digit=2)
```

```
[1] "Confusion Matrix"
```

A matrix: 2 × 2 of type chr

	Prediction: target=0	Prediction: target=1
Actual: target=0	89	49
Actual: target=1	31	134

In [12]:
```
library(pROC)

labels <- heart_data$target
predictions <- logit1$fitted.values

roc <- roc(labels ~ predictions)

print("Area Under the Curve (AUC)")
round(auc(roc),4)

print("ROC Curve")
# True Positive Rate (Sensitivity) and False Positive Rate (1 - Specificity)
plot(roc, legacy.axes = TRUE)
```

```
Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

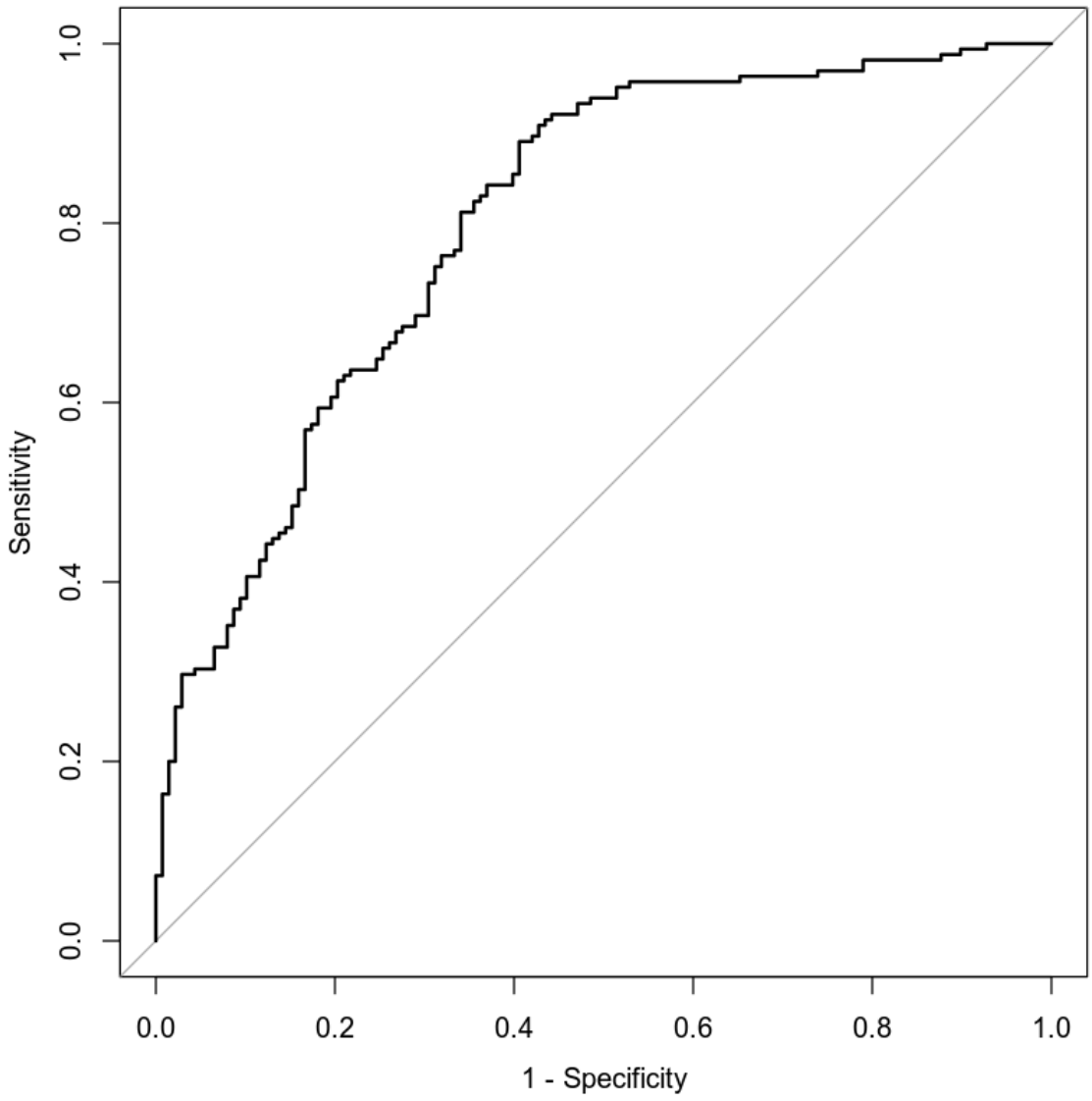
    cov, smooth, var

Setting levels: control = 0, case = 1
Setting direction: controls < cases

[1] "Area Under the Curve (AUC)"
```

0.8007

```
[1] "ROC Curve"
```



In [14]:
```
print("Prediction: 50 years old, resting blood pressure of 122, exercise induc
ed angina, maximum heart rate of 140")
newdata1 <- data.frame(age=50 , trestbps=122 , exang="1" , thalach=140)
pred1 <- predict(logit1, newdata1, type='response')
round(pred1, 4)

print("Prediction: 50 years old, resting blood pressure of 130, no exercise in
duced angina, maximum heart rate of 165")
newdata2 <- data.frame(age=50 , trestbps=130 , exang="0" , thalach=165)
pred2 <- predict(logit1, newdata2, type='response')
round(pred2, 4)
```

```
[1] "Prediction: 50 years old, resting blood pressure of 122, exercise induce
d angina, maximum heart rate of 140"
```

1: 0.2716

```
[1] "Prediction: 50 years old, resting blood pressure of 130, no exercise ind
uced angina, maximum heart rate of 165"
```

1: 0.7853

Model #2 - Second Logistic Regression Model

Logistic regression model for heart disease *(target)* using the variables age of the individual *(age)*, resting blood pressure (trestbps), type of chest pain *(cp)* , maximum heart rate achieved *(thalach)*, as well as the quadratic term for age and the interaction term between age and maximum heart rate achieved.

In [15]:
```
# Create the complete model
logit2 <- glm(target ~ age + trestbps + cp + thalach + I(age^2) + age:thalach,
data = heart_data, family = "binomial")

summary(logit2)
```

```
Call:
glm(formula = target ~ age + trestbps + cp + thalach + I(age^2) +
    age:thalach, family = "binomial", data = heart_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.6961  -0.7537   0.2925   0.7123   2.3058

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.556e+01  1.054e+01  -1.476  0.13988
age          1.744e-01  2.669e-01   0.653  0.51357
trestbps    -1.958e-02  8.978e-03  -2.181  0.02916 *
cp1          1.913e+00  4.437e-01   4.313 1.61e-05 ***
cp2          2.037e+00  3.473e-01   5.867 4.45e-09 ***
cp3          1.777e+00  5.477e-01   3.245  0.00117 **
thalach      1.363e-01  5.119e-02   2.663  0.00775 **
I(age^2)     8.424e-04  1.750e-03   0.481  0.63025
age:thalach -1.867e-03  8.909e-04  -2.095  0.03616 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 417.64  on 302  degrees of freedom
Residual deviance: 293.67  on 294  degrees of freedom
AIC: 311.67

Number of Fisher Scoring iterations: 5
```

In [16]:
```
library(ResourceSelection)
print("Hosmer-Lemeshow Goodness of Fit Test")
hl = hoslem.test(logit2$y, fitted(logit2), g=50)
hl
```

```
[1] "Hosmer-Lemeshow Goodness of Fit Test"

        Hosmer and Lemeshow goodness of fit (GOF) test

data:  logit2$y, fitted(logit2)
X-squared = 52, df = 48, p-value = 0.3209
```

In [17]:
```
default_model_data <- heart_data[c('age', 'trestbps', 'cp', 'thalach')]
pred <- predict(logit2, newdata=default_model_data, type='response')

depvar_pred = as.factor(ifelse(pred >= 0.5, '1', '0'))

# Create the confusion matrix
conf.matrix <- table(heart_data$target, depvar_pred)[c('0','1'),c('0','1')]
rownames(conf.matrix) <- paste("Actual", rownames(conf.matrix), sep = ": targe
t=")
colnames(conf.matrix) <- paste("Prediction", colnames(conf.matrix), sep = ": t
arget=")

# Print nicely formatted confusion matrix
print("Confusion Matrix")
format(conf.matrix,justify="centre",digit=2)
```

```
[1] "Confusion Matrix"
```

A matrix: 2 × 2 of type chr

	Prediction: target=0	**Prediction: target=1**
Actual: target=0	102	36
Actual: target=1	36	129

In [18]:
```
library(pROC)

labels <- heart_data$target
predictions <- logit2$fitted.values

roc <- roc(labels ~ predictions)

print("Area Under the Curve (AUC)")
round(auc(roc),4)

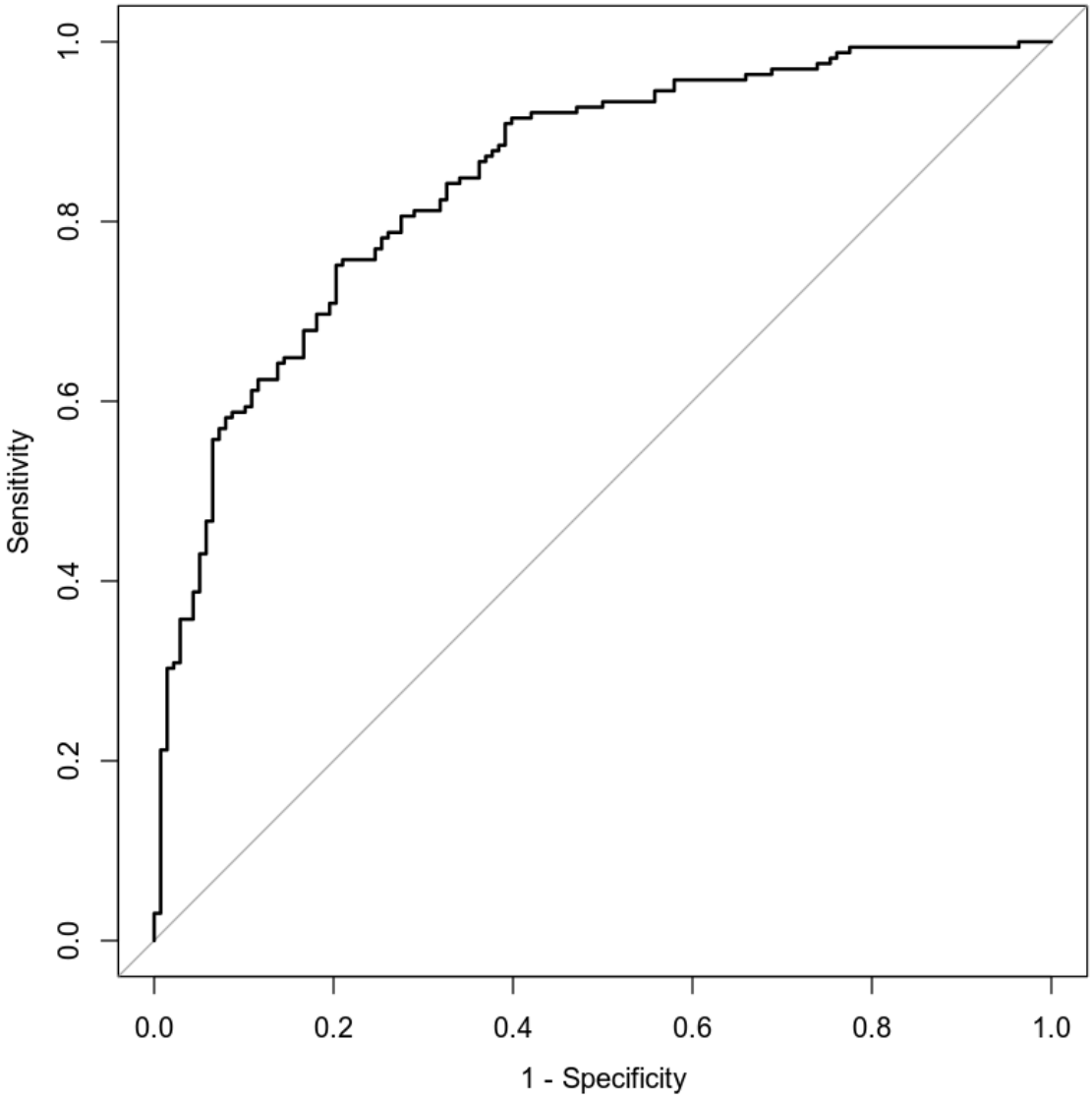
print("ROC Curve")
# True Positive Rate (Sensitivity) and False Positive Rate (1 - Specificity)
plot(roc, legacy.axes = TRUE)
```

```
Setting levels: control = 0, case = 1
Setting direction: controls < cases

[1] "Area Under the Curve (AUC)"
```

0.8478

```
[1] "ROC Curve"
```



In [20]:
```
print("Prediction: 50 years old, resting blood pressure of 115, no chest pain,
maximum heart rate of 133")
newdata1 <- data.frame(age=50 , trestbps=115 , cp="0" , thalach=133)
pred1 <- predict(logit2, newdata1, type='response')
round(pred1, 4)

print("Prediction: 50 years old, resting blood pressure of 125, typical angin
a, maximum heart rate of 155")
newdata2 <- data.frame(age=50 , trestbps=125 , cp="1" , thalach=155)
pred2 <- predict(logit2, newdata2, type='response')
round(pred2, 4)
```

```
[1] "Prediction: 50 years old, resting blood pressure of 115, no chest pain,
maximum heart rate of 133"
```

1: 0.2188

```
[1] "Prediction: 50 years old, resting blood pressure of 125, typical angina,
maximum heart rate of 155"
```

1: 0.8007

Random Forest Classification Model

Random forest classification model for the presence of heart disease *(target)* using the variables age *(age)*, sex *(sex)*, chest pain type *(cp)*, resting blood pressure *(trestbps)*, cholesterol measurement *(chol)*, resting electrocardiographic measurement *(restecg)*, exercise-induced angina *(exang)*, and number of major vessels *(ca)*.

In [1]:
```
heart_data <- read.csv(file="heart_disease.csv", header=TRUE, sep=",")

# Converting appropriate variables to factors
heart_data <- within(heart_data, {
   target <- factor(target)
   sex <- factor(sex)
   cp <- factor(cp)
   fbs <- factor(fbs)
   restecg <- factor(restecg)
   exang <- factor(exang)
   slope <- factor(slope)
   ca <- factor(ca)
   thal <- factor(thal)
})

head(heart_data, 10)

print("Number of variables")
ncol(heart_data)

print("Number of rows")
nrow(heart_data)
```

A data.frame: 10 × 14

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	th
<int>	<fct>	<fct>	<int>	<int>	<fct>	<fct>	<int>	<fct>	<dbl>	<fct>	<fct>	<
62	1	2	130	231	0	1	146	0	1.8	1	3	3
58	0	0	130	197	0	1	131	0	0.6	1	0	2
60	0	3	150	240	0	1	171	0	0.9	2	0	2
63	1	0	140	187	0	0	144	1	4.0	2	2	3
62	1	0	120	267	0	1	99	1	1.8	1	2	3
63	0	2	135	252	0	0	172	0	0.0	2	0	2
43	1	0	150	247	0	1	171	0	1.5	2	0	2
42	1	2	120	240	1	1	194	0	0.8	0	0	3
59	1	2	126	218	1	1	134	0	2.2	1	1	1
48	1	0	124	274	0	0	166	0	0.5	1	0	3

```
[1] "Number of variables"
```

14

```
[1] "Number of rows"
```

303

In [23]:
```
set.seed(6522048)

# Partition the data set into training and testing data
samp.size = floor(0.85*nrow(heart_data))

# Training set
print("Number of rows for the training set")
train_ind = sample(seq_len(nrow(heart_data)), size = samp.size)
train.data1 = heart_data[train_ind,]
nrow(train.data1)

# Testing set
print("Number of rows for the testing set")
test.data1 = heart_data[-train_ind,]
nrow(test.data1)
```

```
[1] "Number of rows for the training set"
```

257

```
[1] "Number of rows for the testing set"
```

46

In [25]:
```
set.seed(6522048)
library(randomForest)

# checking
#=====================================================================
train = c()
test = c()
trees = c()

for(i in seq(from=1, to=150, by=1)) {
    #print(i)

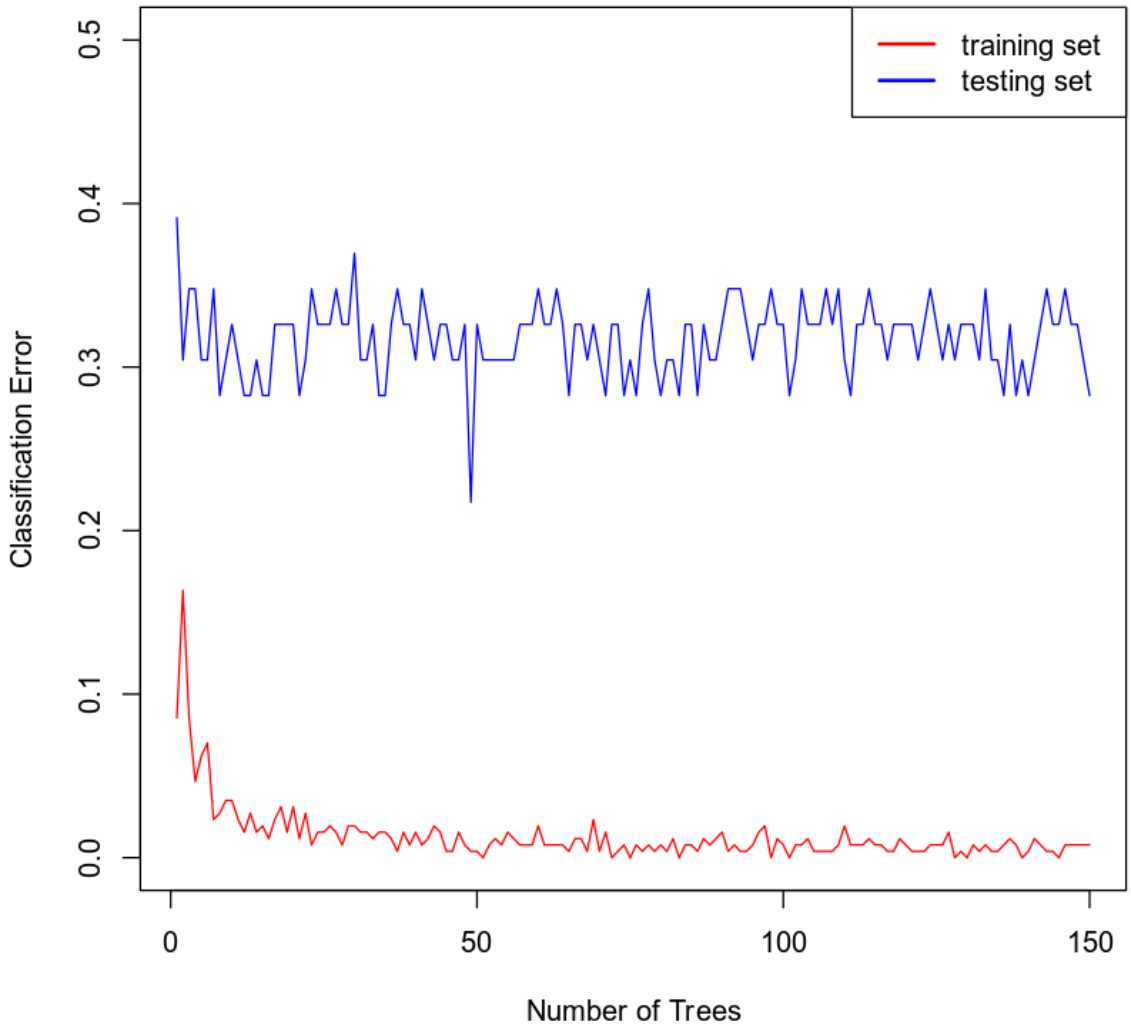
    trees <- c(trees, i)

    model_rf1 <- randomForest(target ~ age + sex + cp + trestbps + chol + rest
ecg + exang + ca, data=train.data1, ntree = i)

    train.data.predict1 <- predict(model_rf1, train.data1, type = "class")
    conf.matrix1 <- table(train.data1$target, train.data.predict1)
    train_error = 1-(sum(diag(conf.matrix1)))/sum(conf.matrix1)
    train <- c(train, train_error)

    test.data.predict1 <- predict(model_rf1, test.data1, type = "class")
    conf.matrix2 <- table(test.data1$target, test.data.predict1)
    test_error = 1-(sum(diag(conf.matrix2)))/sum(conf.matrix2)
    test <- c(test, test_error)
}

plot(trees, train,type = "l",ylim=c(0,0.5),col = "red", xlab = "Number of Tree
s", ylab = "Classification Error")
lines(test, type = "l", col = "blue")
legend('topright',legend = c('training set','testing set'), col = c("red","blu
e"), lwd = 2 )
```


training set
testing set
Classification Error
0.0
0.1
0.2
0.3
0.4
0.5
0
50
100
150
Number of Trees

In [27]:
```
set.seed(6522048)
library(randomForest)
model_rf2 <- randomForest(target ~ age + sex + cp + trestbps + chol + restecg
+ exang + ca, data=train.data1, ntree = 20)

# Confusion matrix
print("============================================================================
=============================================")
print('Confusion Matrix: TRAINING set based on random forest model built using
20 trees')
train.data.predict1 <- predict(model_rf2, train.data1, type = "class")

# Construct the confusion matrix
conf.matrix <- table(train.data1$target, train.data.predict1)[,c('0','1')]
rownames(conf.matrix) <- paste("Actual", rownames(conf.matrix), sep = ": ")
colnames(conf.matrix) <- paste("Prediction", colnames(conf.matrix), sep = ":
")

# Print nicely formatted confusion matrix
format(conf.matrix,justify="centre",digit=2)


print("============================================================================
=============================================")
print('Confusion Matrix: TESTING set based on random forest model built using
20 trees')
test.data.predict1 <- predict(model_rf2, test.data1, type = "class")

# Construct the confusion matrix
conf.matrix <- table(test.data1$target, test.data.predict1)[,c('0','1')]
rownames(conf.matrix) <- paste("Actual", rownames(conf.matrix), sep = ": ")
colnames(conf.matrix) <- paste("Prediction", colnames(conf.matrix), sep = ":
")

# Print nicely formatted confusion matrix
format(conf.matrix,justify="centre",digit=2)
```

```
[1] "============================================================================
============================================="
[1] "Confusion Matrix: TRAINING set based on random forest model built using
20 trees"
```

A matrix: 2 × 2 of type chr

	Prediction: 0	**Prediction: 1**
Actual: 0	118	2
Actual: 1	0	137

```
[1] "============================================================================
============================================="
[1] "Confusion Matrix: TESTING set based on random forest model built using 2
0 trees"
```

A matrix: 2 × 2 of type chr

	Prediction: 0	**Prediction: 1**
Actual: 0	11	7
Actual: 1	8	20

Random Forest Regression Model

Random forest regression model for maximum heart rate achieved using the variables age *(age)*, sex *(sex)*, chest pain type *(cp)*, resting blood pressure *(trestbps)*, cholesterol measurement *(chol)*, resting electrocardiographic measurement *(restecg)*, exercise-induced angina *(exang)*, and number of major vessels *(ca)*.

In [28]:
```
set.seed(6522048)

# Partition the data set into training and testing data
samp.size = floor(0.80*nrow(heart_data))

# Training set
print("Number of rows for the training set")
train_ind = sample(seq_len(nrow(heart_data)), size = samp.size)
train.data2 = heart_data[train_ind,]
nrow(train.data2)

# Testing set
print("Number of rows for the testing set")
test.data2 = heart_data[-train_ind,]
nrow(test.data2)
```

```
[1] "Number of rows for the training set"
```

242

```
[1] "Number of rows for the testing set"
```

61

In [29]:

```
set.seed(6522048)
library(randomForest)

# Root mean squared error
RMSE = function(pred, obs) {
    return(sqrt( sum( (pred - obs)^2 )/length(pred) ) )
}

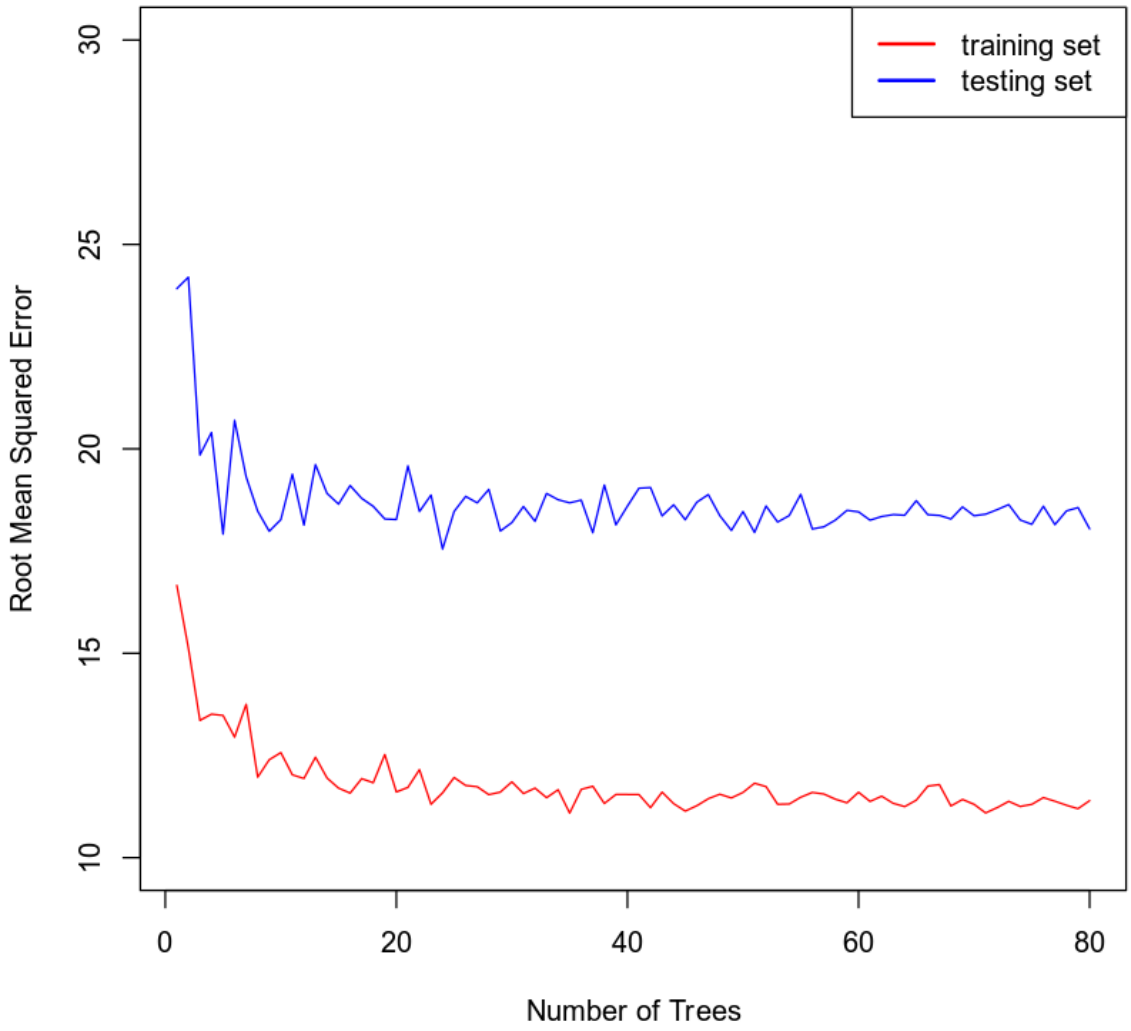

# checking
#=====================================================================
train = c()
test = c()
trees = c()

for(i in seq(from=1, to=80, by=1)) {
    trees <- c(trees, i)
    model_rf3 <- randomForest(thalach ~ age + sex + cp + trestbps + chol + res
tecg + exang + ca, data=train.data2, ntree = i)

    pred <- predict(model_rf3, newdata=train.data2, type='response')
    rmse_train <-  RMSE(pred, train.data2$thalach)
    train <- c(train, rmse_train)

    pred <- predict(model_rf3, newdata=test.data2, type='response')
    rmse_test <-  RMSE(pred, test.data2$thalach)
    test <- c(test, rmse_test)
}

plot(trees, train,type = "l",ylim=c(10,30),col = "red", xlab = "Number of Tree
s", ylab = "Root Mean Squared Error")
lines(test, type = "l", col = "blue")
legend('topright',legend = c('training set','testing set'), col = c("red","blu
e"), lwd = 2 )
```


training set
testing set
Root Mean Squared Error
10
15
20
25
30
0
20
40
60
80
Number of Trees

In [30]:
```
set.seed(6522048)
library(randomForest)
model_rf4 <- randomForest(thalach ~ age + sex + cp + trestbps + chol + restecg
+ exang + ca, data=train.data2, ntree = 20)


# Root mean squared error
RMSE = function(pred, obs) {
    return(sqrt( sum( (pred - obs)^2 )/length(pred) ) )
}

print("================================================================================
================================================")
print('Root Mean Squared Error: TRAINING set based on random forest model buil
t using 20 trees')
pred <- predict(model_rf4, newdata=train.data2, type='response')
RMSE(pred, train.data2$thalach)


print("================================================================================
================================================")
print('Root Mean Squared Error: TESTING set based on random forest model built
using 20 trees')
pred <- predict(model_rf4, newdata=test.data2, type='response')
RMSE(pred, test.data2$thalach)
```

```
[1] "================================================================================
================================================"
[1] "Root Mean Squared Error: TRAINING set based on random forest model built
using 20 trees"
```

11.7762979145166

```
[1] "================================================================================
================================================"
[1] "Root Mean Squared Error: TESTING set based on random forest model built
using 20 trees"
```

18.7266873327815